

Qatar-4b

R-2677

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_170203 · created 2026-08-30T23:09:12+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457787.61821 +/- 0.00087 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.053
Transit depth (Rp/Rs)^2	3.09 +/- 1.87 [%]
Orbital Inclination (inc)	85.68 +/- 2.13
Airmass coefficient 1 (a1)	1.104 +/- 0.068
Airmass coefficient 2 (a2)	-0.054 +/- 0.054
Scatter in the residuals of the lightcurve fit is	6.03 %
Best Comparison Star	#1 - [315, 58]
Optimal Aperture	4.55
Optimal Annulus	11.45
Transit Duration (day)	0.0364 +/- 0.0077

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.176 $\pm$ 0.053 vs 0.138 (deviation 0.72 σ)
Duration fit vs published	0.0364 d vs 0.0892 d (deviation 6.85 σ)
Mid-time O-C	-3.1 min
Depth SNR	3.3
Residual scatter	6.03 %

Light curve

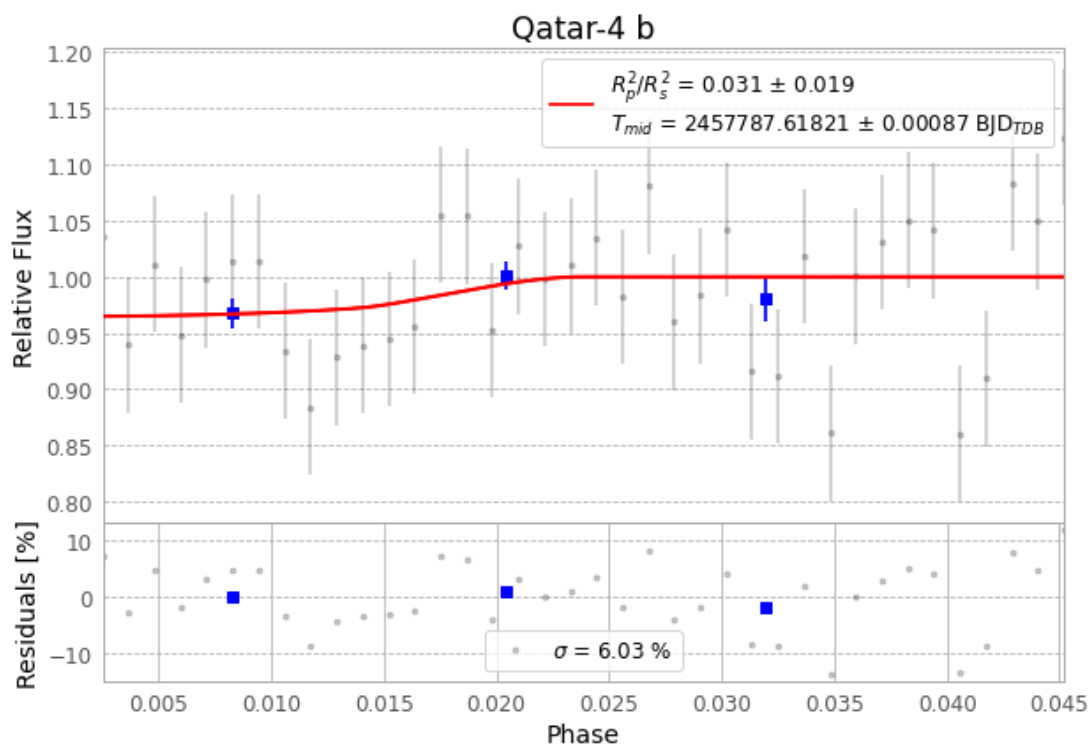


Figure 1 — FinalLightCurve_Qatar-4 b_2017-02-02.png (SHA-256 9794523e561d9087...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [204, 269], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ee8ed60323de028b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

62 FITS frames (41 MB), Qatar-4170203025712.FITS → Qatar-4170203060012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-170203.log (c3a9f8864e56...), FinalLightCurve_Qatar-4 b_2017-02-02.png (9794523e561d...), FinalLightCurve_Qatar-4 b_2017-02-02.csv (8ac0b703ffe7...)

Compute resources

Wall time 190.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 23:09Z.